Experiment No: 7

AIM: Implement the embedding wallet (Metamask) and transaction using Solidity

Theory:

What is MetaMask?
MetaMask is a popular cryptocurrency wallet and browser extension that enables users to interact with blockchain networks, especially Ethereum. It acts as a bridge between web browsers and decentralized applications (DApps), allowing users to access blockchain services directly from their browser.
MetaMask manages Ethereum accounts by securely storing public and private keys. It allows users to perform operations such as sending and receiving Ether (ETH), interacting with smart contracts, and signing transactions digitally. It also provides a user-friendly interface to manage multiple accounts and switch between different blockchain networks such as Ethereum Mainnet and test networks like Sepolia.
One of the key features of MetaMask is transaction signing. Whenever a user performs an action (like deploying a smart contract or voting in a ballot), MetaMask prompts the user to confirm the transaction, ensuring security and user control. It also calculates gas fees required to execute transactions on the blockchain.
Security is an important aspect of MetaMask. It uses a secret recovery phrase (seed phrase) to restore wallets, and users must keep this phrase confidential. If lost, the wallet cannot be recovered.
In summary, MetaMask is an essential tool for blockchain development and usage, acting as both a secure wallet and a gateway to decentralized applications.

What is a Testnet?
A testnet is a blockchain network that is used exclusively for testing and development. It replicates the functionality of a real blockchain (mainnet) but uses free and non-valuable cryptocurrency. This allows developers to experiment with smart contracts and transactions without risking real money.
Testnets are important because blockchain transactions are irreversible and may involve costs. By using a testnet, developers can debug errors, optimize smart contracts, and ensure correct functionality before deploying on the mainnet.
Testnets simulate real blockchain conditions such as:

- Transaction validation
- Gas fees
- Network delays
- Smart contract execution

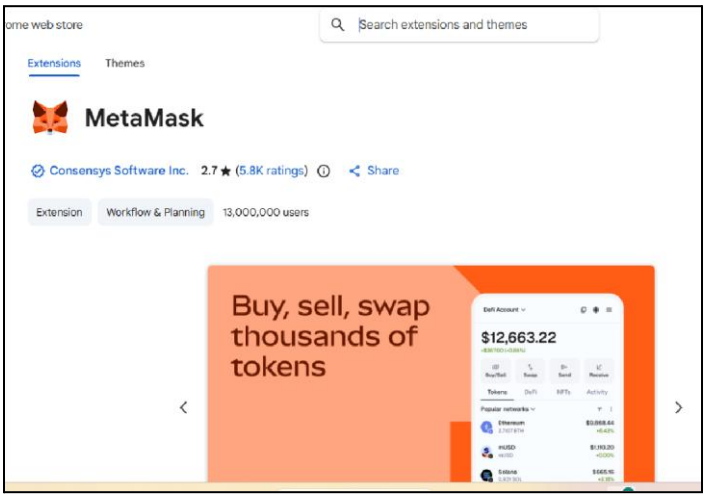
Developers obtain test Ether (ETH) from faucets, which are online services that distribute free tokens for testing purposes. Popular Ethereum testnets include Sepolia and Goerli (now deprecated).
Using testnets improves reliability, reduces financial risk, and helps developers gain confidence before launching applications on the main network.

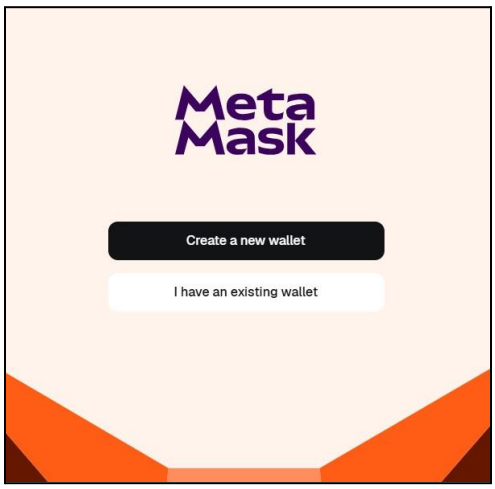
Steps to Connect MetaMask with Remix IDE for Performing Transactions

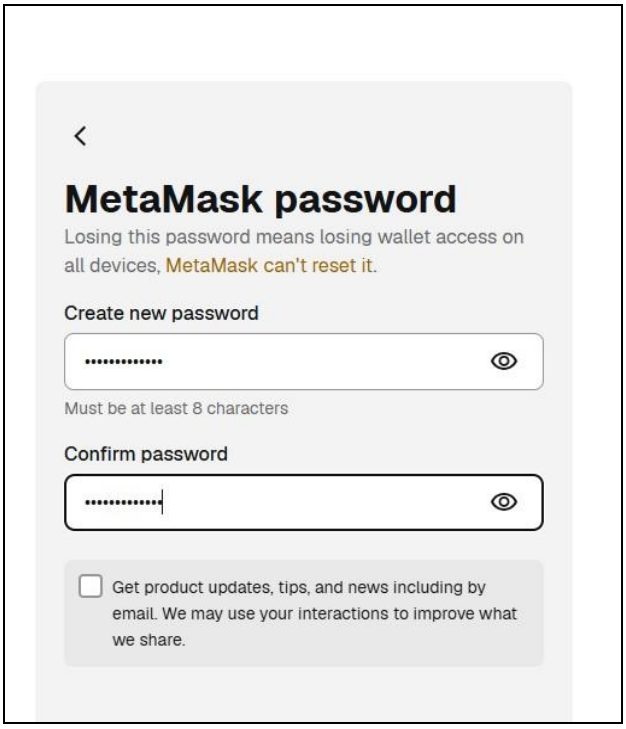
1. Install the MetaMask extension in your browser and create a new wallet by setting a password and saving the secret recovery phrase securely.
2. Open Remix IDE in your web browser.
3. Write or paste your smart contract code in Remix and compile it using a suitable Solidity compiler version.
4. Navigate to the “Deploy & Run Transactions” tab in Remix.
5. In the Environment dropdown, select “Injected Provider - MetaMask”.
6. A MetaMask popup will appear requesting permission; click “Connect” to link Remix with your wallet.
7. In MetaMask, choose a test network such as Sepolia to perform safe testing.
8. Obtain test Ether (ETH) from a faucet to pay for gas fees required for transactions.
9. Click the “Deploy” button in Remix; MetaMask will display a transaction confirmation window.
10. Review gas fees and contract details, then click “Confirm” to deploy the contract.
11. Once deployed, the contract will appear under the “Deployed Contracts” section in Remix.
12. Interact with the contract functions such as vote, delegate, or giveRightToVote. Each action will trigger a MetaMask confirmation request.
13. Confirm transactions in MetaMask to execute them on the blockchain

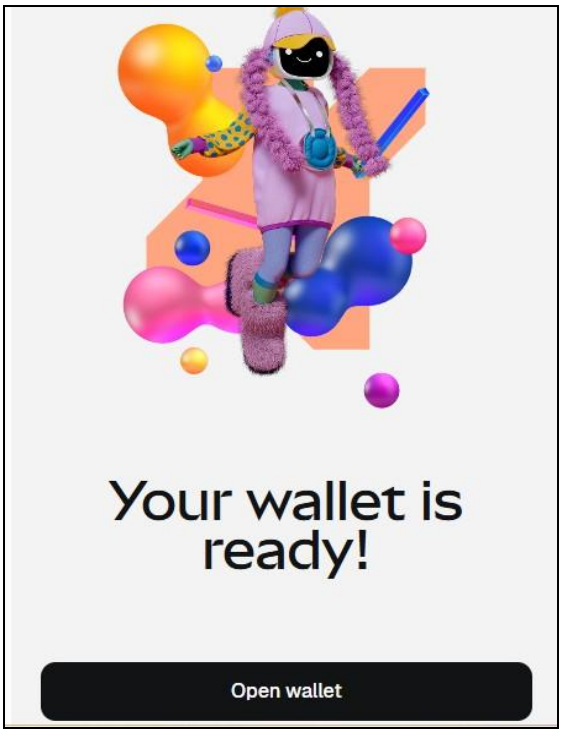
Implementation

Step 1: Install MetaMask
Install MetaMask extension and create a wallet

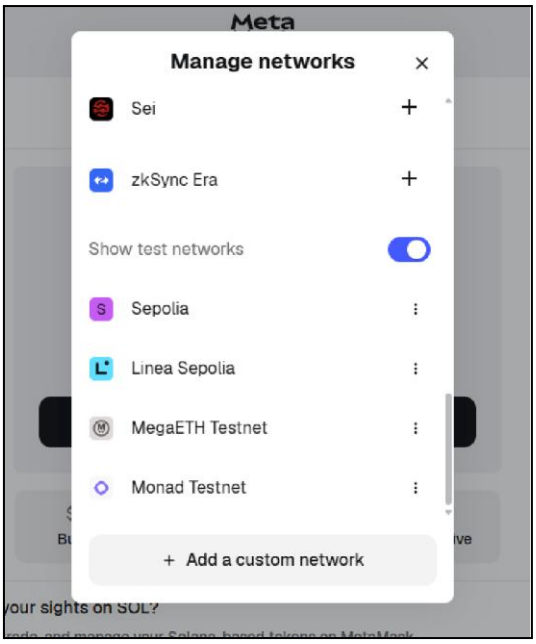


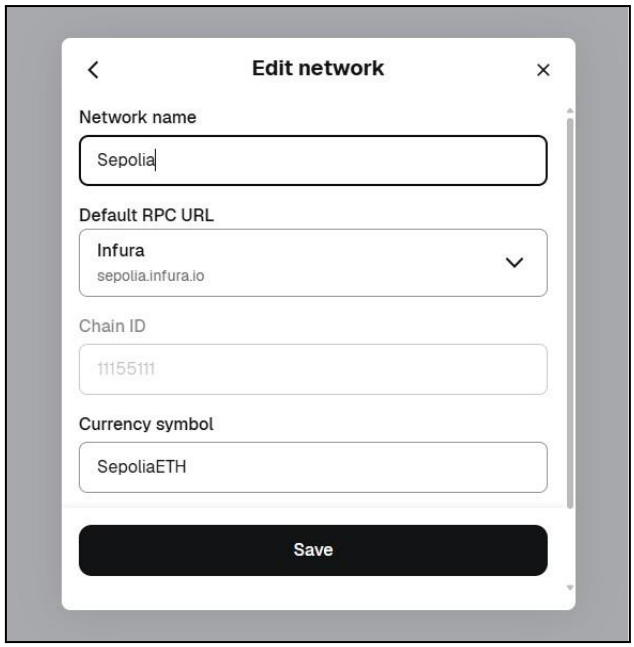


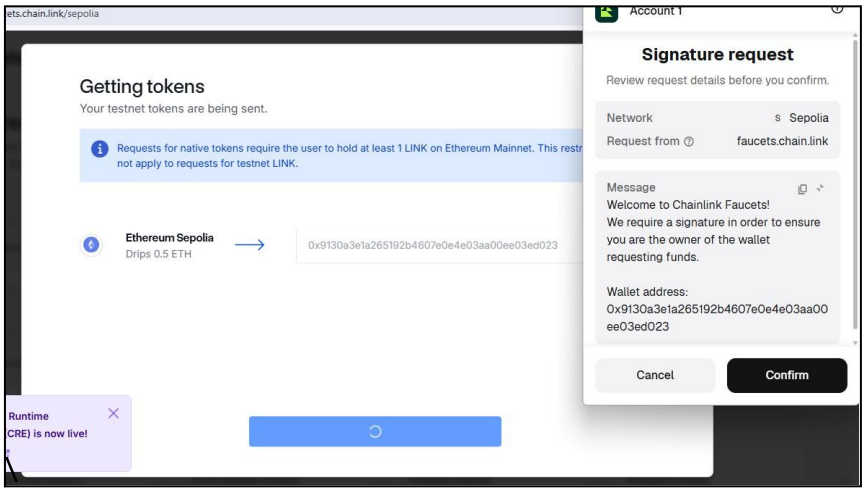


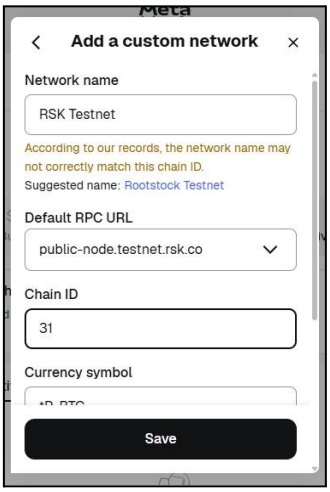


Step 2: Add Sepolia Testnet
Configure MetaMask to connect to Sepolia network.


Meta
Manage networks
Sei
zkSync Era
Show test networks
Sepolia
Linea Sepolia
MegaETH Testnet
Monad Testnet
+ Add a custom network

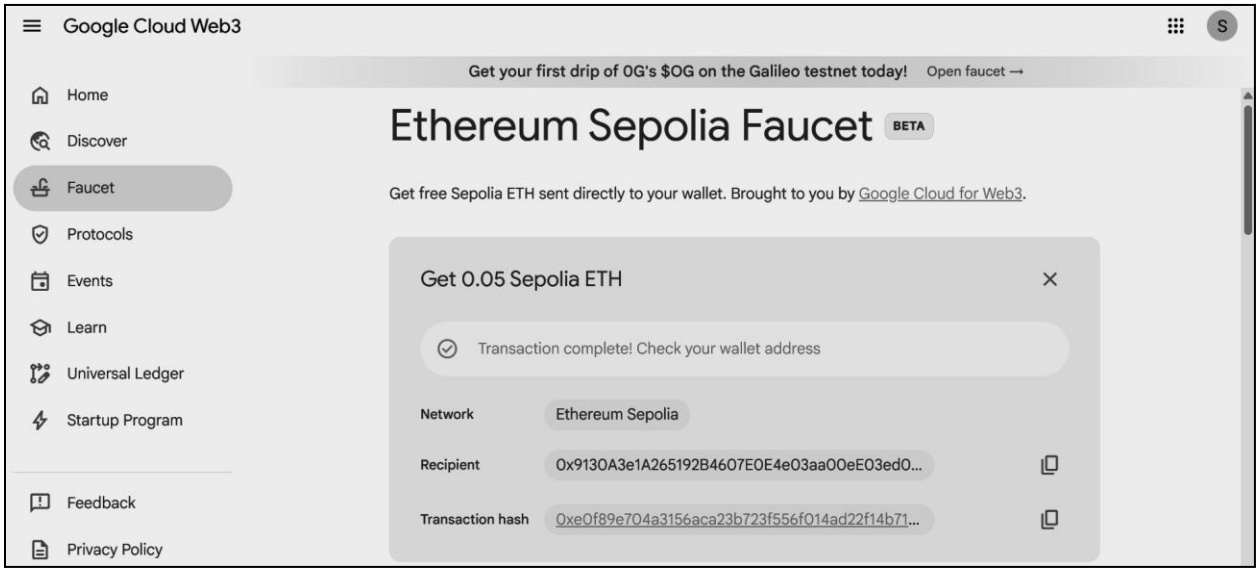

Edit network
Network name
Sepolia
Default RPC URL
Infura
sepolia.infura.io
Chain ID
11155111
Currency symbol
SepoliaETH
Save


ets.chain.link/sepolia
Getting tokens
Your testnet tokens are being sent.
Requests for native tokens require the user to hold at least 1 LINK on Ethereum Mainnet. This restr
not apply to requests for testnet LINK.
Ethereum Sepolia
Drips 0.5 ETH
0x9130a3e1a265192b4607e0e4e03aa00ee03ed023
Runtime
CRE) is now live!
Account 1
Signature request
Review request details before you confirm.
Network
Sepolia
Request from
faucets.chain.link
Message
Welcome to Chainlink Faucets!
We require a signature in order to ensure you are the owner of the wallet requesting funds.
Wallet address:
0x9130a3e1a265192b4607e0e4e03aa00ee03ed023
Cancel
Confirm


Add a custom network
Network name
RSK Testnet
According to our records, the network name may not correctly match this chain ID.
Suggested name: Rootstock Testnet
Default RPC URL
public-node.testnet.rsk.co
Chain ID
31
Currency symbol
Save

Step 3: Fund Wallet

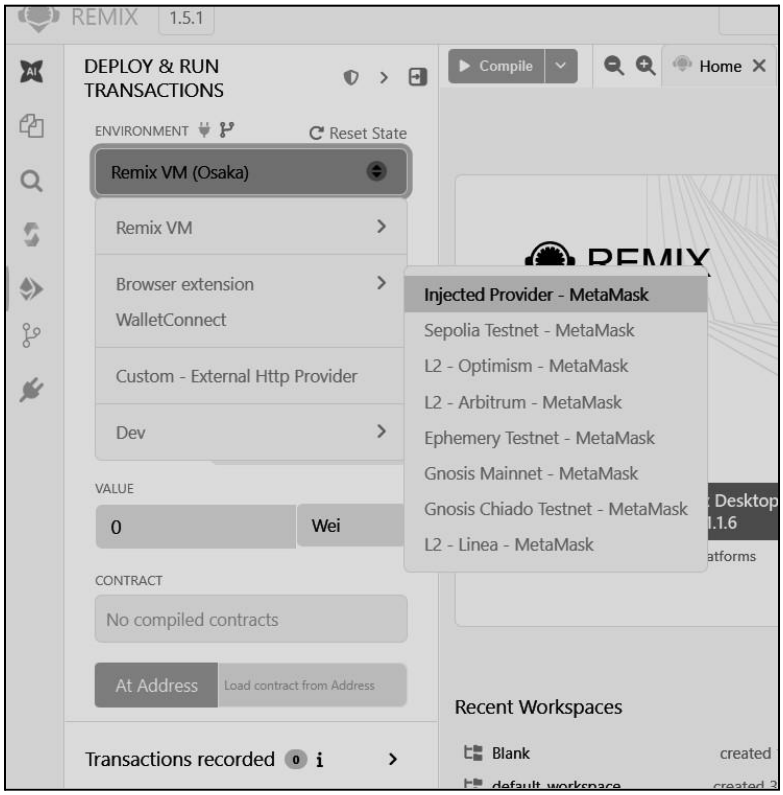
Use faucet to get test ETH

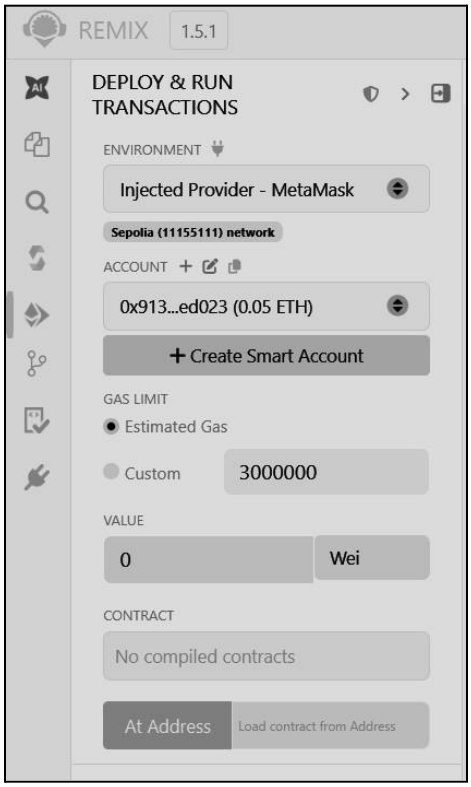


Step4: Connect the Sepolia Testnet / RSK Testnet to Remix IDE

1. In the Remix IDE, select the Environment as Injected Web3.

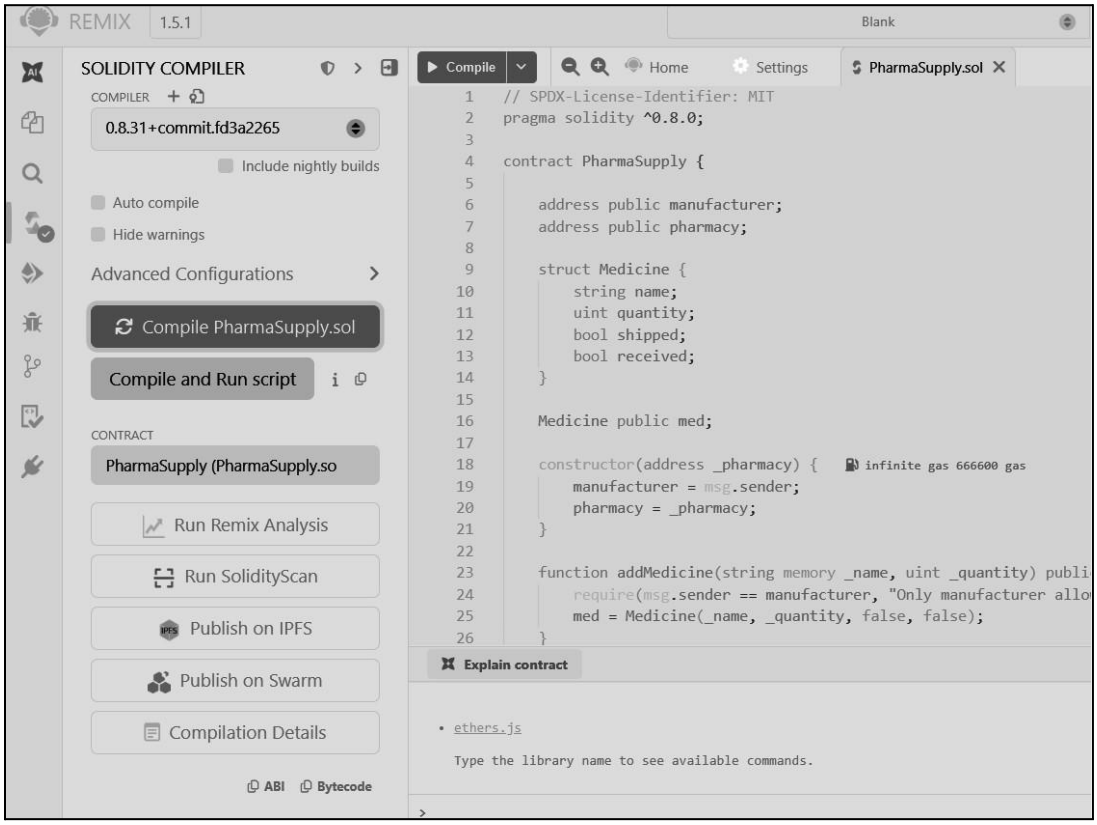
Note : Injected Web3 connects Remix with the active account in Metamask.





Compile the Contract

1. Open Remix
2. Paste your contract code
3. Go to Solidity Compiler (left panel)
4. Select version 0.8.x (match ^0.8.0)
5. Click Compile PharmaSupply.sol

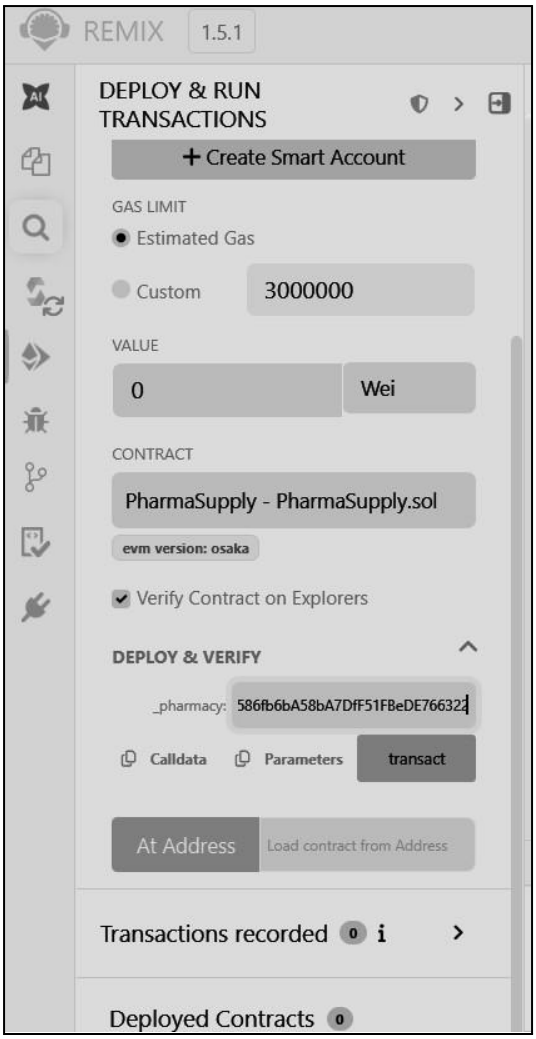


Connect MetaMask (Injected Provider)

1. Go to Deploy & Run Transactions tab
2. In Environment → select Injected Provider - MetaMask
3. MetaMask popup will open → click Connect

Make sure:

- Network = Sepolia Test Network
- You have 0.05 Sepolia ETH



Deploy Contract

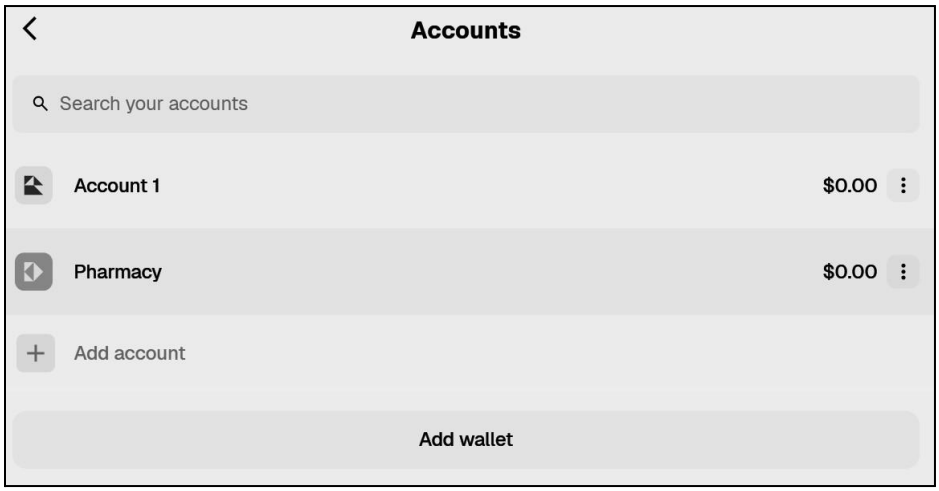
Your constructor needs a pharmacy address

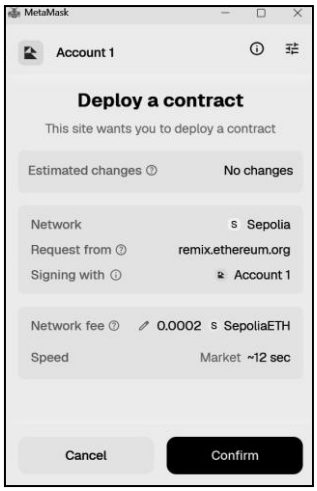
Steps:

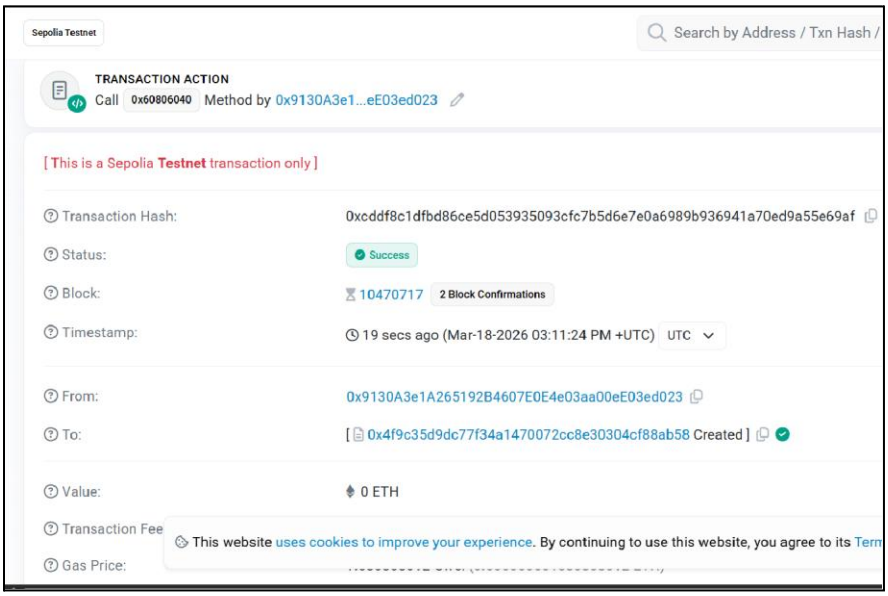
1. Copy your second MetaMask account address (pharmacy)
   ○ MetaMask → Account 2 → Copy address
2. Paste that address in constructor field
3. Click Deploy
4. MetaMask will pop up → Confirm transaction

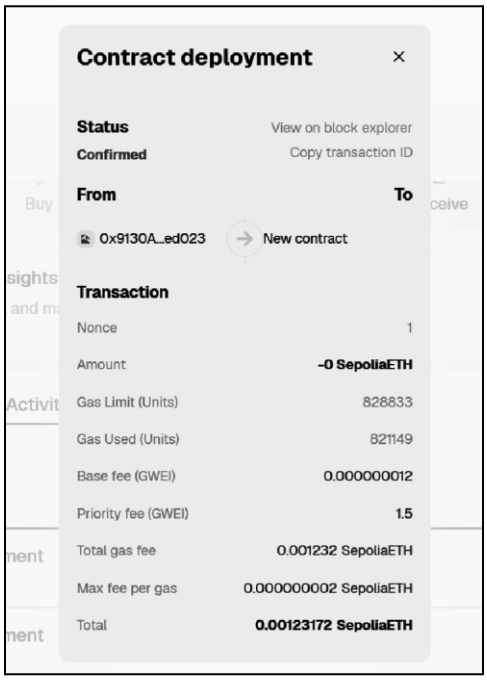
Wait a few seconds…

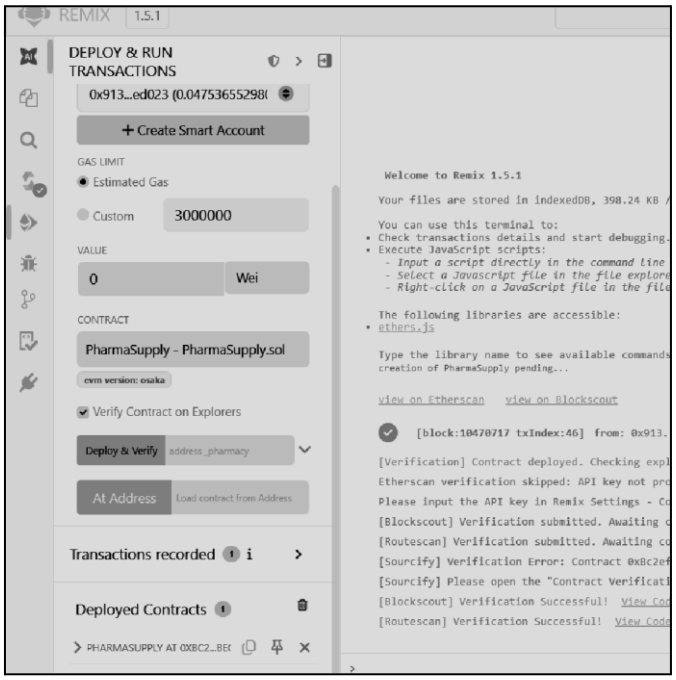
After success → your contract appears under Deployed Contracts


Accounts
Search your accounts
Account 1
$0.00
Pharmacy
$0.00
Add account
Add wallet


MetaMask
Account 1
Deploy a contract
This site wants you to deploy a contract
Estimated changes
No changes
Network
Sepolia
Request from
remix.ethereum.org
Signing with
Account 1
Network fee
0.0002 SepoliaETH
Speed
Market ~12 sec
Cancel
Confirm


Sepolia Testnet
Search by Address / Txn Hash /
TRANSACTION ACTION
Call 0x60806040 Method by 0x9130A3e1...eE03ed023
[This is a Sepolia Testnet transaction only]
Transaction Hash:
0xcddf8c1dfbd86ce5d053935093cfc7b5d6e7e0a6989b936941a70ed9a55e69af
Status:
Success
Block:
10470717
2 Block Confirmations
Timestamp:
19 secs ago (Mar-18-2026 03:11:24 PM +UTC)
UTC
From:
0x9130A3e1A265192B4607E0E4e03aa00eE03ed023
To:
[0x4f9c35d9dc77f34a1470072cc8e30304cf88ab58 Created]
Value:
0 ETH
Transaction Fee
Gas Price:
This website uses cookies to improve your experience. By continuing to use this website, you agree to its Term


Contract deployment
Status
Confirmed
View on block explorer
Copy transaction ID
From
To
0x9130A...ed023
New contract
Transaction
Nonce
1
Amount
-0 SepoliaETH
Gas Limit (Units)
828833
Gas Used (Units)
821149
Base fee (GWEI)
0.000000012
Priority fee (GWEI)
1.5
Total gas fee
0.001232 SepoliaETH
Max fee per gas
0.000000002 SepoliaETH
Total
0.00123172 SepoliaETH

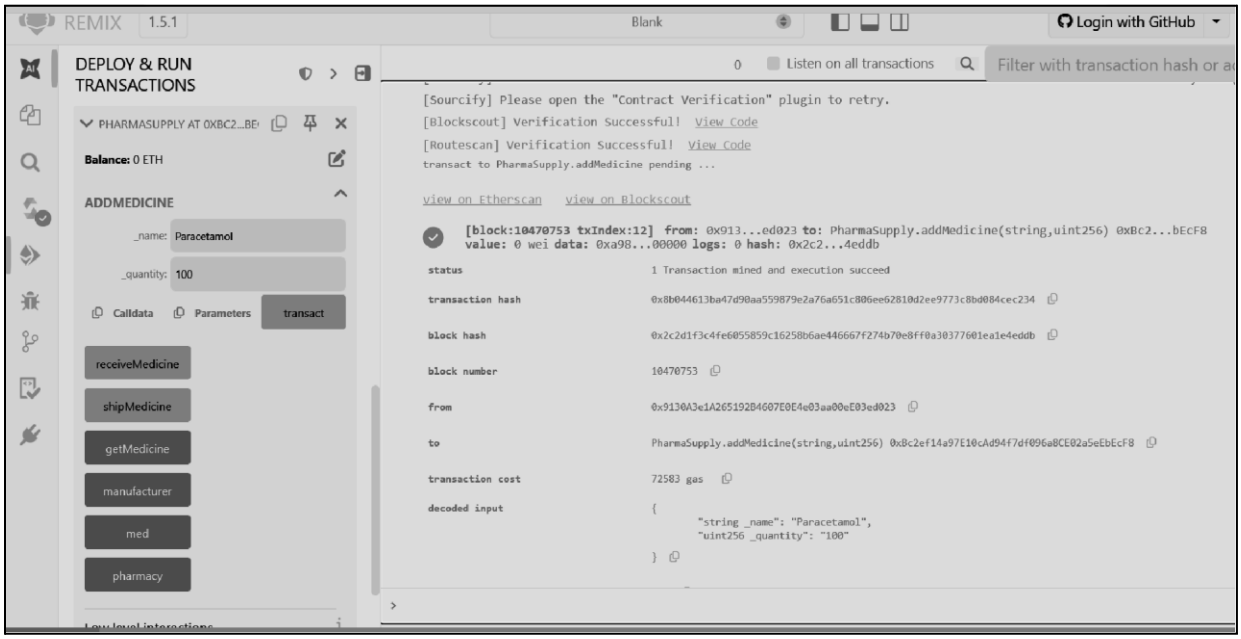

REMIX 1.5.1
DEPLOY & RUN TRANSACTIONS
+ Create Smart Account
GAS LIMIT
Estimated Gas
Custom
3000000
VALUE
0
Wei
CONTRACT
PharmaSupply - PharmaSupply.sol
Verify Contract on Explorers
Deploy & Verify
At Address
Transactions recorded
Deployed Contracts
Welcome to Remix 1.5.1

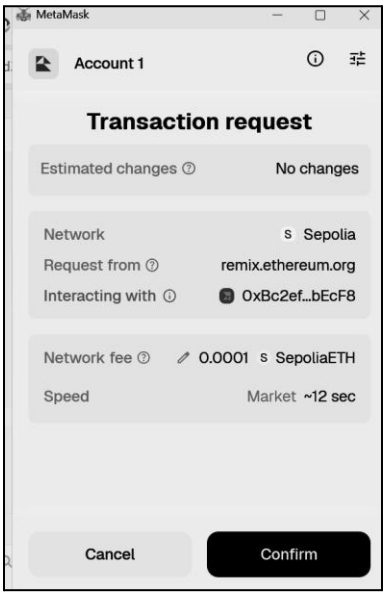
Interact with Contract

Step 1: Add Medicine (Manufacturer only)

- Make sure MetaMask is on Account 1 (manufacturer) ● In Remix:
  - Enter:
    - _name: "Paracetamol"
    - _quantity: 100
- Click addMedicine

Confirm in MetaMask

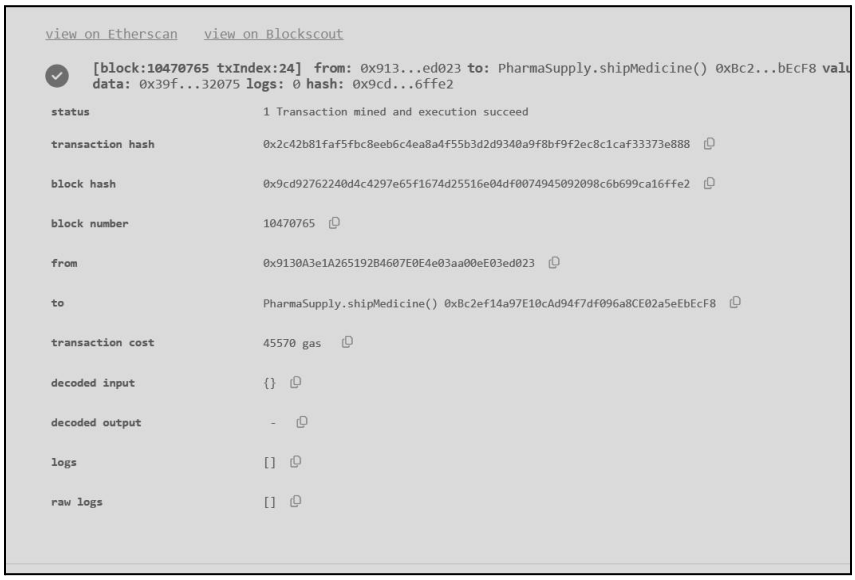


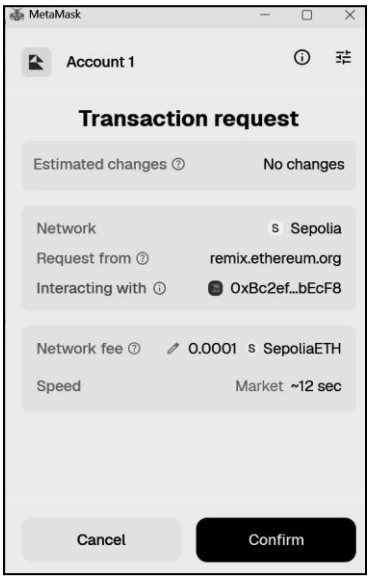


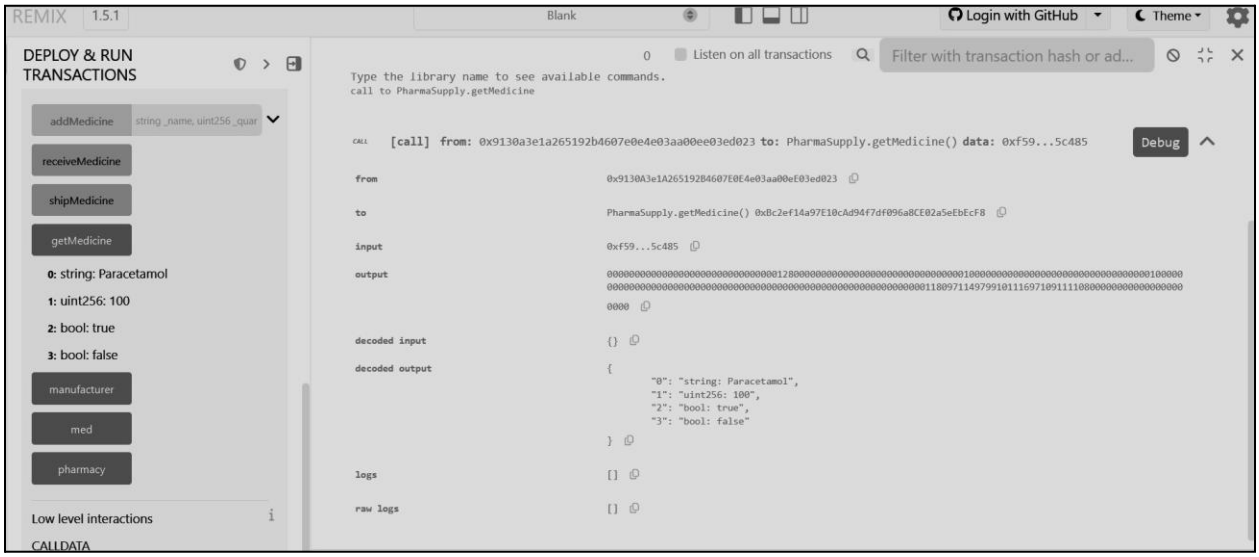
Step 2: Ship Medicine

- Still on Account 1
- Click shipMedicine

- Confirm transaction



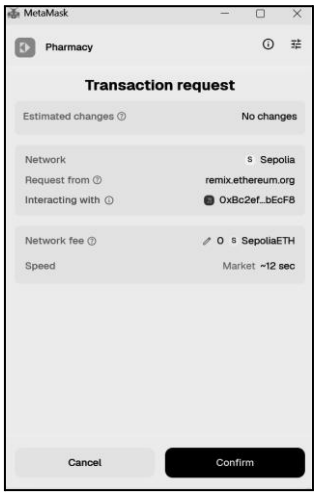


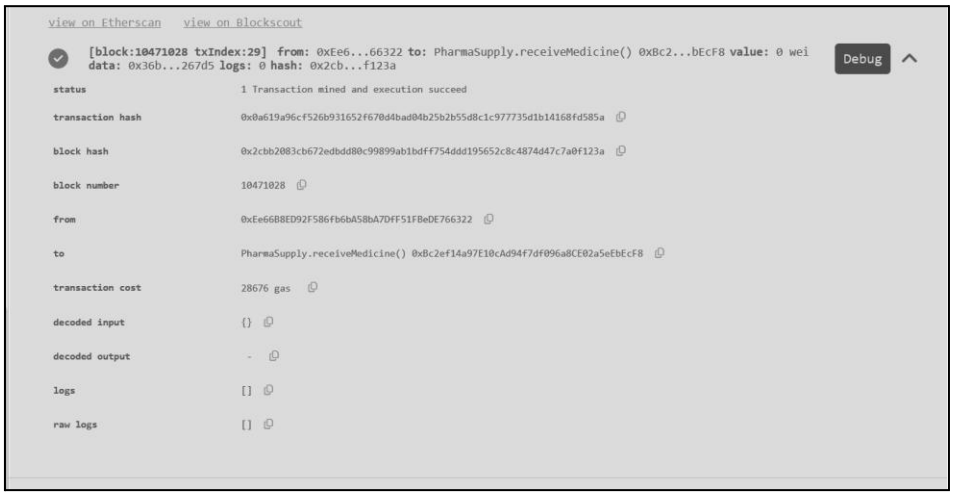


Step 3: Receive Medicine (Pharmacy)

- Switch MetaMask to Account 2 (pharmacy)
- Click receiveMedicine

- Confirm transaction



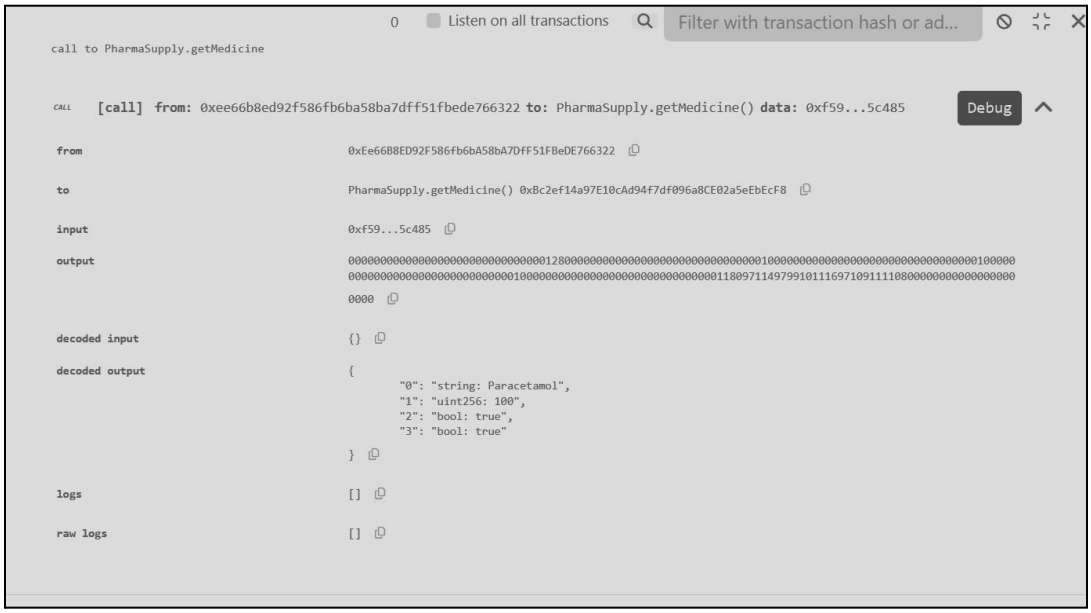


Step 4: Check Data

- Click getMedicine

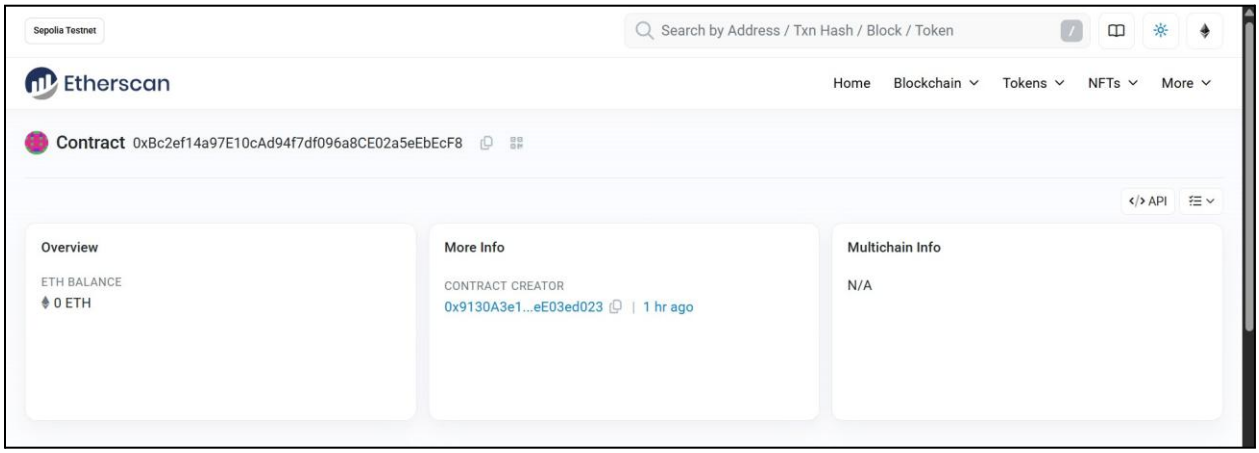
Output will be like:

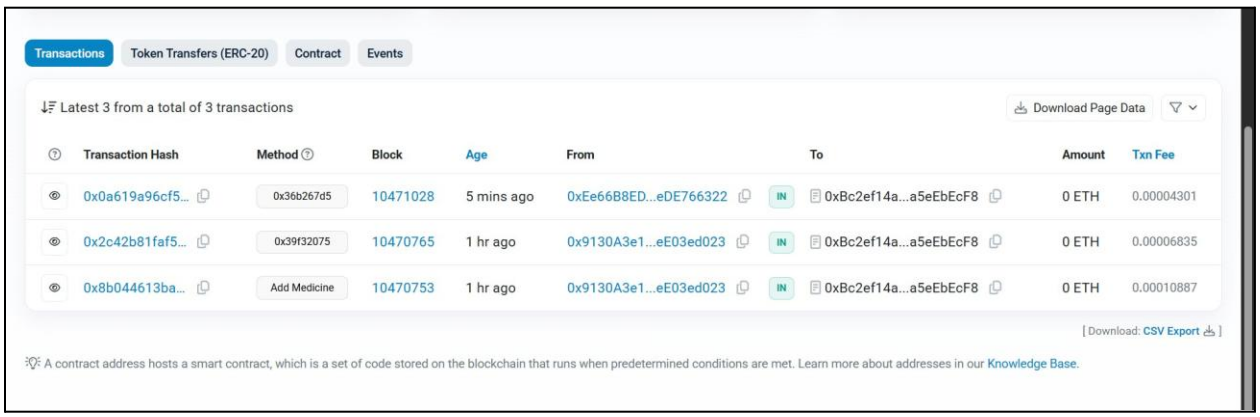
"Paracetamol"
100 true
(shipped)
true (received)



Compiled PharmaSupply.sol

Copy contract address → paste in Etherscan (Sepolia) See all transactions + status





Conclusion

The experiment successfully demonstrated MetaMask integration with Remix IDE and execution of blockchain transactions. It provided hands-on experience with smart contracts, ensuring secure and transparent interactions on a testnet